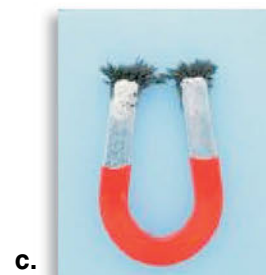
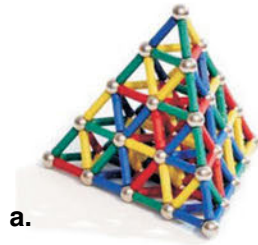


UNIT 7

Magnetism and the world we live in

A. What do the pictures a, b and c have in common?

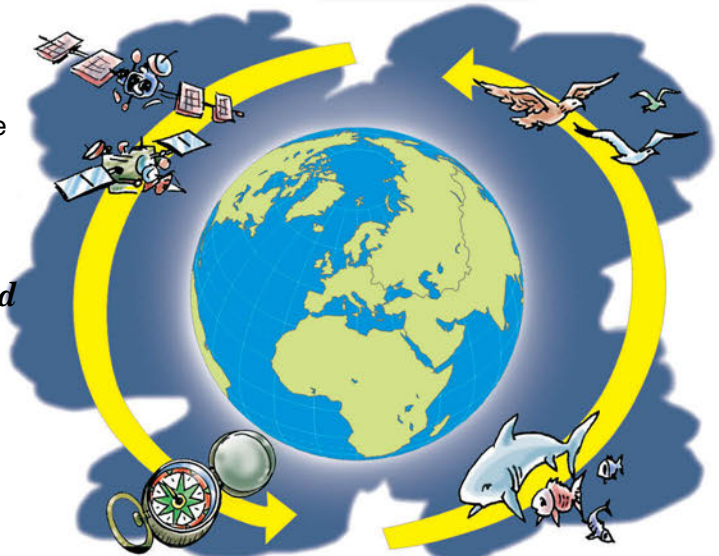
- i) they are all made of plastic
- ii) they involve magnetism
- iii) they are about enjoyment.



B. Discuss the following questions:

- i) Why don't the pieces of the pyramid fall?
- ii) How do the roller coaster cars stay on the track?
- iii) How does the item in picture 'c' work?

C. Look at the sketch of the world and say how magnetic fields influence our world. Make a list and compare it with your class.



Grammar:

Past Continuous
Used to
Past Continuous vs. Past Simple

Functions:

Narrating an event from the past

Vocabulary:

Science and explanations

Learning strategies:

When I want to remember new words I...

- associate new words with similar words in Greek
- imagine the words in a context
- put the word in a phrase or sentence
- repeat the word to myself in my room

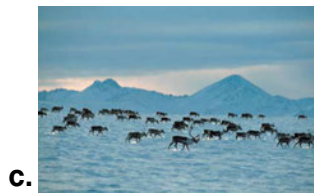
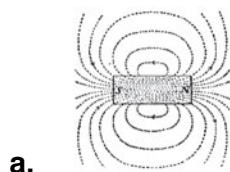
Lesson 1

AIMS

- To listen for implied information
- To guess the main story from headlines

Lead-in

A. Match the pictures with the words in the box.



migrating animals

magnetic field

magnetism

B. Look at the picture on the right. What do you think the people are doing? Why?



Listening

A. Listen to the conversations between some teenagers and find out what the mystery in the picture is. You can find the answer written backwards below.



remmus hcae htron dna retniw hcae htuos ylf dna etargim yeht nehws msitengam esu sdrib

B. Listen to the children's conversation again and answer the following questions. You can also read the cartoon script in the Resource Material on p. 158.

1. Which of the boys probably lived in another country when he was small? What does he say that tells you the answer?
2. What did the old men in the village use to tell the boys about the birds?
3. Where were the birds going for the winter?

C. Think of some examples of people leaving where they live or moving around the world. For example, Africans taken to America for slavery; the Kurds in Iraq; moving to another city because of a job transfer; having to move because of a natural disaster.

Who?
When?
Where?
Why?

Use the words in the box to ask and answer questions about the migration of people.

Bring a photo to the class of a member of your family who has emigrated to another country. Tell his/her story to the class. Use the questions in the box to tell your story.



Grammar

Past Continuous

In pairs, find an example of something that was happening in the past in the cartoon script in the Resource Material on p. 158 in the Appendix.

Then use the dialogue examples to help you complete the rules below with the missing words and circle the correct word in italics.

Rule 1: We use **was/were** + **verb** + _____ when we talk about a *continuous* action in the past.

Rule 2: We use the **time** word _____ when we want to talk about an action that was happening at the same time as another action in the past.

Usage rules:

In pairs, circle the correct word in the following rules.

When is used to say that something *never/always* happens or happened in particular circumstances.

While is used to *combine/ separate* a continuous action in the past with a specific action.

Used to

*In pairs, circle the verb **used to** in the cartoon script on p.158. Then complete the rules about when we use **used to**.*

Rule 1: **Used to** is used to say that something happened *regularly /rarely* in the past but _____ happen now.

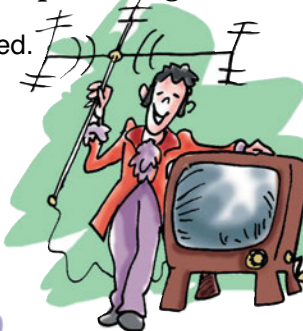
Rule 2: **Used to** describes *past / present* situations.

Game: Fact or Myth?

Task 1

In pairs, decide which of the following statements is a fact or a myth. Ask your physics teacher or do a web-search on the internet to check your answers. You can also look up the names at <http://www.wikipedia.org/>.

- i) Einstein invented the fridge but the gas leaked.
- ii) Planck discovered electricity.
- iii) Newton invented the television.
- iv) Maxwell discovered magnetism.
- v) Galileo invented the thermometer.



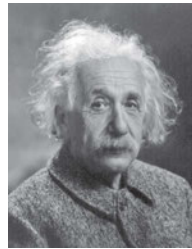
Lesson 1

Task 2

A. Match the names of the scientists in the box with the pictures below. Use the dates to help you. Then match the scientists with the statements 1-5 below.



a 1858-1947



b 1879-1955



c 1642-1727



d 1831-1879



e 1564-1642

Galileo
Newton
Planck
Maxwell
Einstein

- 1) He used to work in the Patent Office in Zurich before he developed $E=Mc^2$. He did not agree with using the atomic bomb.
- 2) An Italian scientist who used to sing songs to measure short intervals of time. He proved that the earth revolves around the sun.
- 3) He used to have the nickname 'Dafty' Max when he was at school in Edinburgh. He described the laws of electricity and magnetism.
- 4) He used to teach in Cambridge before he moved to London in 1696. He invented the reflecting telescope in 1668.
- 5) He discovered that energy of electromagnetic waves consists of small packets. He used to wear his glasses in bed.

B. Listen to the guide in the Planetarium describing the life and work of the scientists and check your answers.

Why were these scientists important in world development? Find out how their work changed the world. Use the words in the word bank to help you.

WORD BANK
atomic power
improved telescope
laws of motion
light-bulbs
electric fields
magnetic fields

Task 3

Use photos of the scientist you admire most and present them in class. Tell the class where he used to work, where he spent his life, why he became famous.



Reading: Facts or Myths? Lead-in

A. Look at the pictures below. What do you think they have to do with magnetism?

Clues:

What did sailors use in the past to find out which direction they were travelling in?
a) the sun b) the wind c) compass



B. What do you think the story for the following headlines is? Discuss your answers.

The Philadelphia Experiment

Turtles coming home

Plane disappears near Bermuda

C. Now read the short text below to check your answer for the ship.

The **Philadelphia Experiment** was a secret magnetic experiment by the American Navy on October 28th, 1943. Scientists used magnetic fields to make a ship called *The Eldridge* disappear. The American government said that the experiment did not happen but sailors on the ship said it did.

D. In small groups discuss if you think this story is real or if it is just a myth.

Task 1

Read the two texts below quickly and decide which of them is about:

- i) magnetism ii) migration

Text A

Almost everyone who has used a compass knows that our planet has a magnetic field. On the earth's surface, it is weaker than the typical magnet you find on your fridge door. But even today, most people regard magnetism as a mystery which we know very little about. We see the affect of magnets in our everyday lives. It is common to see a television or computer screen shaking when a mobile phone rings next to it.

Text B

Scientists used to believe that animals used their instinct to migrate but now they know that they also use magnetic fields. Although it is a mystery, many scientists now believe that animals like turtles or the simple pigeon have a magnetic sense which they use for navigation to find their way. Cells in an animal's brain contain magnetite, an iron oxide crystal that aligns with magnetic north similar to a compass needle. This guides them when they migrate.

Task 2

A. Tell your partner if you think magnetism played a role in each of these stories. Why?

B. Read the two texts again carefully and discuss with your partner which text mentions: a) the negative effects of magnetic fields, b) the positive uses of magnetic fields, c) an electrical appliance in our kitchens.

C. In pairs, decide which of the texts mentions mysterious behaviour.

In **Appendix I (IT'S YOUR CHOICE)** you can find another article about a sea mystery to do with magnetism.

Task 3

Magnetic tapes (such as the VHS tape or a music cassette) use magnetism to record sound or pictures.

Look around your house and make a list of all the items that operate because of magnetism. Ask your Physics teacher for information about the magnetic materials used in different household items.



Lesson 2

AIMS

- To raise awareness of word origins
- To listen and label a diagram



Vocabulary

Task 1

A. Use your dictionary to make a word tree for each of the words: *monotony*, *aeroplane*, *microscope*. In pairs, complete the table:

-graph	(write)	Photograph
bi-	(two)	Bicycle
		Monotony
		Aeroplane
		Microscope

Can you guess what scientific words come from *Magnesia* and *Magnes*?

B. Match the words in the box to the origins.

physics araba
Spartan kutu
pullover aerobic
house mystery
irgat hooligan
kasap sandwich

Greek	Turkish	English

C. Read the text by Xenophon Zolotas (26/9/1957) in Appendix II (p.159) and circle all the Greek words in the text. Then, in pairs, write a sentence in English with 10 of these words. Compare your answers with the rest of the class.

Task 2 - Useful Words

A. Use your dictionary to help you complete the definition of the words in bold with words from the box.

attract
away
closer
compass
flow
force
iron
needle
poles
repel

magnetism: the (i)..... of substances like (ii)..... or some other metals to (iii)..... or (iv)..... each other because of electric charges.

compass needle: a (v)..... which moves according to the (vi)..... of the magnet field. The (vii)..... needle tells us where north and south (viii)..... are.

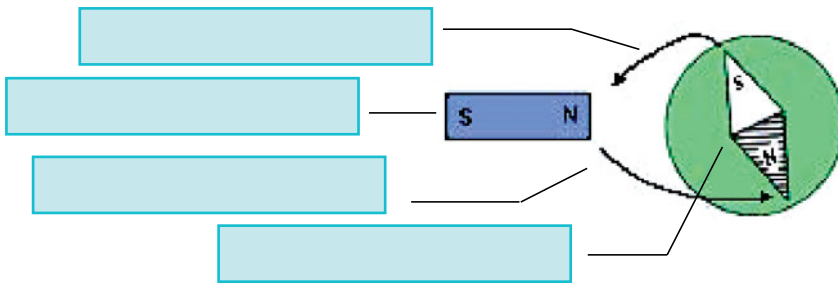
attract: to make something come (ix)

repel: to push something (x).....

Magnetic fields

B. In pairs, label the diagram with the words:

magnet, attract, repel, compass needle.



C. In pairs, look at the items on the right. Discuss how you can use these items to make a compass. Use your compass to find north. In which direction is your house from your classroom?



Listening &

Listening 1 - Magnetic Fields

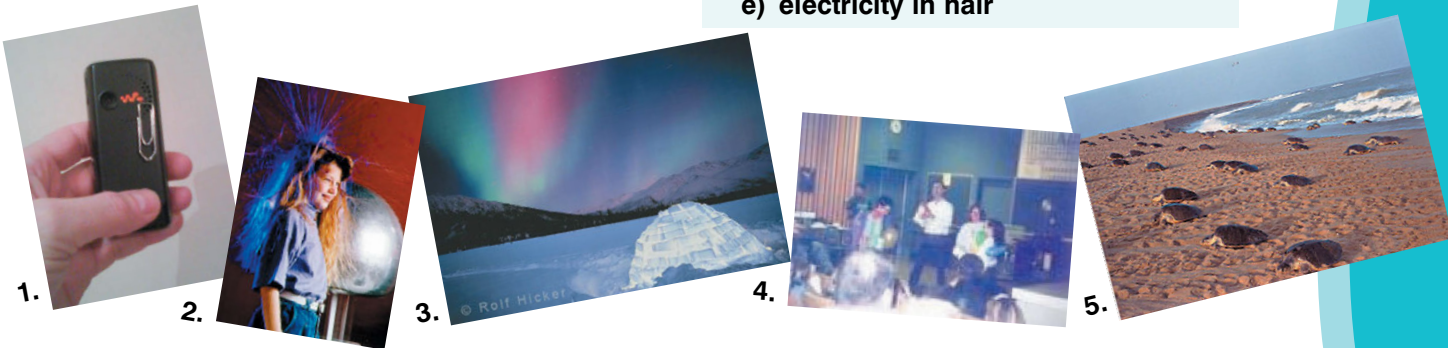


Speaking

Task 1

A. Match the list of natural phenomena a-e in the box to the pictures 1-5 below.

- a) magnetic fields in mobile phones
- b) the Aurora Borealis
- c) magnetic balloons
- d) migrating turtles
- e) electricity in hair



B In small groups discuss what causes hair to stand up in the morning when people get out of bed. Think of ways to control this static electricity in people's hair.

Task 2 - Static Electricity

Listen to the lecture on static electricity and tick the statements as TRUE or FALSE. Use the information in the table to give advice to your partner about static electricity.

	TRUE	FALSE
1. Because similar charges repel, every hair wants to leave your head.		
2. The northern lights are the result of two magnetic fields.		
3. When you separate newly washed clothes that are stuck, they make a popping sound.		
4. The air rubbed around a balloon feels strange.		
5. You can sometimes make a spark when you touch a doorknob.		

Lesson 2

Task 3 - A song

Listen to a song and, with your partner, decide what the song is about. Is the topic of the song a myth or a fact? How do you know?

What does she imply by some of the lyrics in the song?



Listening &



Speaking

Sherlock Holmes and Dr. Watson are having a holiday in Cephalonia. They heard about the mystery of the missing turtle eggs on a beach nearby and decided to help the locals.

Task 1

A. Listen to the story and make notes about what each person was doing that day and try to find out who stole the eggs.

B. Role Play

The Missing Eggs

In groups of four take roles for

i) Sherlock Holmes, ii) the Spanish tourist, iii) the stranger, and iv) Freddie.

You can use the sketches for ideas.

Sherlock asks questions like: What were YOU doing when...?



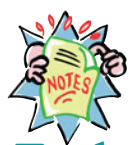
Speaking - Retelling a story

Look at the photographs below. In pairs, discuss what you can tell from each photograph about the people, the clothes, the items, the period in history.

In pairs, create a short story using the four pictures. Share your story with the class.



Tip: Think of **when, where, what, who, why, how.**



Writing

Task 1

Discuss in pairs how you think myths begin. For example: was the story of Heracles a myth or a fact or part myth part fact?

Task 2

In pairs, look at the speech bubbles from the sailors of a ship that found the Marie Celeste and find the differences between what the sailors said and the newspaper. Then compare your answers and report to the class.



a

When I entered the captain's cabin, I saw two cups of coffee sitting on his desk

b

While I was looking in the hold, I saw that the cargo was oranges and fruit

c

The crew's boots and jackets were lying on the deck but everything else was in perfect condition

d

The stove was still warm and so I think the cook was preparing lunch when they disappeared

e

The compass needle was working perfectly crazy and pointed North

f

Some ropes were hanging over the side of the ship. I counted them twice to double check

g

The ship's log said that the ship was heading for Italy

New York Herald

Feb 26th, 1873

The Ghost Ship: the Mystery of the Mary Celeste

According to the captain's log, the ship was heading for Spain when the mystery occurred. It was carrying a cargo of alcohol. The captain liked to drink whiskey and this was probably the cause of the mystery. Reports from the first sailors

on board said that there was no food on the ship and that everything was a mess. They said that clothes and books were lying on the deck. Many ropes were hanging over the side of the ship. Another sailor reported that the compass was working properly.

Task 3 - Story writing competition

In groups of three, find an unusual story that is reported in the news. Find two newspapers which report the same story. Each member of the group reads a different newspaper. Compare the stories in the newspapers to find the differences and similarities and then combine the facts from them to write your own account of the story for the school newspaper. Present it as a group to the rest of the class.

Lesson 3

Save the turtle

Task 1

Look at the map on the right. In which part of Greece is this place?

In groups of three, decide why the following are a problem for migrating turtles laying eggs.

beach parties sun beds seabirds
tourists fishermen



Source: <http://www.archelon.gr/en>

Task 2

Look at the photographs and decide which pictures show problems and which ones show things that people can do to help the turtles. Add any other ideas you have to help the turtles.



Task 3

In your group, decide if the pictures are good for a poster to help save the turtles. Why? Why not?



What photographs might be better for this project? Decide with your group and find photographs to make people more aware of the problems turtles face.

In what ways could we help the turtles? Make a list of your ideas to help save the turtles then read the ideas below to compare your answers.

Task 4

In your group read the information in the leaflet on 'How you can help'. Decide as a group how you can make a visual to add to a poster for each point in the box to attract people to help the turtles.

- Adopt a sea turtle
- Buy turtle products
- Make a donation of €5 each year
- Become a volunteer to write articles
- Get friends to sponsor a turtle
- Tell friends about the problem

Task 5

Make copies of the sponsor form in Appendix, p. 159 and go round your friends and family and ask them to sponsor a turtle. Use your poster to explain to your friends and family the problems turtles face and the solutions.

Each group then reports back to class how much money they have raised from sponsors.

Activity A

Put the verbs in parentheses into the past simple or past continuous.

While we (have)..... a physics lesson yesterday, the teacher (tell)..... us that it was Einstein who (invent)..... the fridge. We (think)..... he (joke)..... and that this was a myth. But then he (show)..... us a page from a web site which (prove)..... that this was not a myth but a fact. The site said that Einstein invented the fridge as he (try)..... to help his student Leo Szilard to make some money. When they (try)..... the fridge, there were problems because the gas (leak).....

___/5 points

Activity B

Complete the following sentences with 'invent' or 'discover'.

- a) Newton the telescope.
- b) No one has if the Philadelphia Experiment was fact or myth.
- c) Alfred Nobel dynamite in 1866.
- d) Max Planck the quantum nature of energy in 1899.
- e) Marconi how to use electromagnetic waves to send radio signals.

___/5 points

Activity C

Match the following halves of the dialogues.

- | | |
|--|------------------------------------|
| i) George used to like History. | a) They do now, though. |
| ii) Is it true that Newton used to play the violin? | b) No, we know they use magnetism. |
| iii) Did you use to live in Berlin? | c) No, Munich, actually. |
| iv) People didn't use to know much about magnetism. | d) But now he prefers physics. |
| v) People used to think animals used the stars to migrate. | e) Yes, and the piano too. |

___/2.5 points

Self-evaluation

Activity D

Complete the following sentences with words from the box.

migrate
immigrate
emigrate
immigrants
emigrants

- Thousands of Greeks went to live in America in the 1960s.
- There are many who have come to live in Greece.
- Many animals and birds each year in search of food or to lay their eggs.
- There are different reasons why people from their country.
- Some people from other countries have decided to into Greece in search of work.

___/5 points

Activity E

Match the pairs of words in the two lists:

- | | |
|--------------|------------|
| i. compass | a) eggs |
| ii. iron | b) field |
| iii. radio | c) needle |
| iv. magnetic | d) message |
| v. turtle | e) ore |



___/2.5 points

Now tick how well you can do the following:

	With difficulty	Quite well	Easily
✓ I can read and understand texts related to magnetism or two versions of the same incident and find differences	☐	☐	☐
✓ I can listen to and understand scientific talks and stories	☐	☐	☐
✓ I can talk about past situations	☐	☐	☐
✓ I can write a report of an incident using facts	☐	☐	☐